

A Double Counterexample to TLMC Conjecture 00000001190: Prime-Order Elements of $\text{GL}_2(\mathbb{F}_7)$

TLMC Submission — Disproof of Conjecture 00000001190

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Abstract

We refute Conjecture 00000001190, which asserts that the number N_p of elements of prime order p in $\text{GL}_n(\mathbb{F}_q)$ equals $\sum_{r|n, r \text{ prime}} \Theta_r(q)$ with $\Theta_r(q) = q^{n^2-r}(q-1)^{-1} \prod_{i=1}^{r-1} (q^{n-i} - 1)$, “with zero error.” At the single parameter point $(n, q) = (2, 7)$ the formula is wrong twice: it predicts 49 elements of order 2 where the truth is 57, and it predicts 0 elements of order 3 where the truth is 170. Every ground-truth number is a brute-force enumeration of all $7^4 = 2401$ matrices over $\mathbb{Z}/7\mathbb{Z}$, independently machine-verified in Python and by a zero-axiom Lean 4 kernel check.

1 The conjecture

Definition. N_p is the number of elements of prime order p in $\text{GL}_n(\mathbb{F}_q)$.

Conjecture. $N_p = \sum_{r|n, r \text{ prime}} \Theta_r(q)$, where

$$\Theta_r(q) = q^{n^2-r} (q-1)^{-1} \prod_{i=1}^{r-1} (q^{n-i} - 1);$$

this formula covers all prime-order elements with zero error.

2 The formula side at $(n, q) = (2, 7)$

Since $n = 2$, the only prime divisor of n is $r = 2$. Hence the conjectured sum has exactly one term:

$$N_2 \stackrel{!}{=} \Theta_2(7) = 7^{4-2} \cdot \frac{7^{2-1} - 1}{7-1} = 49 \cdot \frac{6}{6} = \boxed{49}, \tag{1}$$

$$N_3 \stackrel{!}{=} 0 \quad (\text{the sum over prime } r \mid 2 \text{ contains no } r = 3, \text{ so it is the empty sum}), \tag{2}$$

$$N_p \stackrel{!}{=} 0 \quad \text{for every prime } p \geq 3. \tag{3}$$

3 Ground truth: enumeration of $\text{GL}_2(\mathbb{F}_7)$

First,

$$|\text{GL}_2(\mathbb{F}_7)| = (7^2 - 1)(7^2 - 7) = 48 \cdot 42 = 2016.$$

This was confirmed by enumerating all $7^4 = 2401$ matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d \in \mathbb{Z}/7\mathbb{Z}$ and keeping exactly those with $ad - bc \not\equiv 0 \pmod{7}$; the count is 2016. Elements of order exactly k are then

quantity	brute-force enumeration	conjectured formula
$ \mathrm{GL}_2(\mathbb{F}_7) $	2016	—
elements of order exactly 2 (N_2)	57	49
elements of order exactly 3 (N_3)	170	0

Table 1: The conjecture fails twice at $(n, q) = (2, 7)$: $57 \neq 49$ and $170 \neq 0$.

those M with $M \neq I$, $M^k = I$ (and automatically $M^j \neq I$ for $1 \leq j < k$ since k is prime). The exhaustive enumeration gives:

As a sanity check, the full order distribution of the 2016 elements is 1:1, 2:57, 3:170, 4:42, 6:618, 7:48, 8:84, 12:84, 14:48, 16:168, 21:96, 24:168, 42:96, 48:336, which sums to 2016.

4 Why the formula undercounts

4.1 Order 2: 57 against 49

Over $\mathbb{F}_7$, $x^2 - 1 = (x - 1)(x + 1)$ splits completely, so every involution is diagonalizable. The elements with eigenvalues $\{1, -1\}$ ($\neq \pm I$) form a single conjugacy class: the centralizer of such an element is the group of diagonal matrices, of order $(7 - 1)^2 = 36$, so the class has size $2016/36 = 56$. Together with the scalar matrix $-I$ (which also has order exactly 2) this gives

$$N_2 = 56 + 1 = 57 \neq 49 = \Theta_2(7).$$

The term $\Theta_2(q) = q^{n^2-2}(q^1 - 1)/(q - 1) = q^{n^2-2} = q^2$ counts the regular-semisimple split-torus contribution (pairs of distinct eigenvalues in $\mathbb{F}_q^\times$, modulo the multiplicative structure of the torus), but it accounts neither for the scalar involution $-I$ nor for the fact that a 2-dimensional representation can realize order 2 with eigenvalue set $\{1, -1\}$ embedded in the conjugation structure differently than the formula assumes.

4.2 Order 3: 170 against 0

In $\mathbb{F}_7^\times$ (cyclic of order 6) there are two primitive cube roots of unity: $x^3 - 1 = (x - 1)(x - 2)(x - 4) \pmod{7}$, indeed $2^3 = 8 \equiv 1$ and $4^3 = 64 \equiv 1 \pmod{7}$. An element of order exactly 3 is diagonalizable with eigenvalues in $\{1, 2, 4\}$, not both 1. The three eigenvalue multisets $\{2, 1\}$, $\{4, 1\}$, $\{2, 4\}$ give three conjugacy classes, each with centralizer of order 36, i.e. 56 elements each; plus the two scalar matrices $2I$ and $4I$. Hence

$$N_3 = 3 \cdot 56 + 2 = 170 \neq 0.$$

The conjecture’s divisor condition “ $r \mid n$ ” is blind to this: because $3 \mid (q - 1) = 6$, the field $\mathbb{F}_7$ itself contains elements of order 3, and scalars λI of order 3 exist in *every* dimension $n \geq 1$, as do diagonalizable ones for all $n \geq 2$. No sum of the form $\sum_{r \mid n} \Theta_r$ can ever count them unless every prime divisor of $q - 1$ happens to divide n .

5 Machine verification

All counts are verified by two independent, axiom-free computations:

- `reproduce.py`: pure-Python enumeration of all 2401 matrices; prints and asserts 2016, 57, 170, 49, 0.
- `lean4/`: Lean 4 (toolchain v4.33.1) theorems proved by `decide` over the same exhaustive enumeration: `count_order2 : ...= 57`, `count_order3 : ...= 170`, `formula2 :  $\Theta_2(7) = 49$` , and `refute :  $57 \neq 49 \wedge 170 \neq 0$` . `#print axioms` confirms every theorem “does not depend on any axioms” (no sorry, no `Classical.choice`, no gaps).

6 Scope and boundaries

We refute only the literal, universal statement of Conjecture 00000001190 at the parameter point $(n, q) = (2, 7)$, where it fails for two different primes p simultaneously. We make no claim about the behaviour of the formula at other parameters; it may be exact or near-exact elsewhere (e.g. when $\gcd(n, q-1) = 1$ the scalar-matrix obstruction disappears, though the 57-vs-49 discrepancy at $p = 2$ here suggests the semisimple counting in Θ_r itself also needs care). Fixing the conjecture would require at least (i) adding scalar contributions $\sum_{\lambda \in \mathbb{F}_q^\times, \text{ord}(\lambda)=p} 1$, and (ii) handling non-regular semisimple classes whenever $p \mid q-1$, independently of any divisibility of n by p .